great deal of experience, or finally, by a scientifically trained metereologist who in addition to the local data also has access to the accurate information supplied by a great many weather-maps from near and far. The forecast of the experienced farmer is more reliable than that of the amateur, and the forecast of the trained metereologist more reliable than that of either of those two. In view of this circumstance, it seems natural to suppose that an ideal intellect, intimately familiar with the most minute details of physical processes occuring concurrently everywhere, would be able to predict tomorrow's weather in all its details with an absolute accuracy. The same idea applies to every other prediction of physical occurrences.

Such an assumption signifies an extrapolation which cannot be demonstrated logically, though it cannot be refused in an *a priori* fashion either; it must therefore be judged not on the score of its truth, but only on the basis of its value. In consequence the actual impossibility of predicting even a single occurrence accurately in classical as well as in quantum physics, appears to be a natural consequence of the circumstance that man with his sense organs and measuring instruments is himself a part of nature, subject to its laws and confined within its limits, whereas the ideal intellect is free of all such limitations.

However, in order to be able to follow through such a view logically, we must comply with an important requirement: We must be on our guard against the temptation to make the ideal intellect the object of a scientific analysis, to regard it as something analogous to ourselves, and to ask of it how it obtains the knowledge which enables it to make precise predictions. The inquisitive human being who would do so, is quite likely to hear this answer to his question: "You resemble the intellect which you comprehend, not *me!*" And if after this reprimand he persists in his obstinacy and declares the concept of an ideal intellect to be meaningless and unnecessary, if not illogical, let him be reminded that not all statements which lack a logical foundation are scientifically worthless, and that his short-sighted formalism stops up the very fountain at which a Galileo, a Kepler, a Newton, and many other great physicists have quenched their thirst for scientific knowledge and insight. For all these men devotion to science was, consciously or unconsciously, a matter of faith—a matter of a serene faith in a rational world order.

Of course, this faith can no more be forced upon anybody than could one be commanded to see the truth or forbidden to commit an error. But the plain fact that we are able, at least to a certain degree, to subject future natural occurrences to our thought processes and to bend them to our will, would be a totally incomprehensible mystery, did it not permit us to surmise at least a certain harmony between the external world and the human intellect. The depth to which one conceives this harmony to extend is a matter of merely secondary importance.